Entity-Relationship Modelling: Concept Rubric

1. Entity and Entity Set

An entity is a real-world object or thing that can be clearly identified and about which the database stores data, such as a student, a course or a lecturer. An entity set is the collection of all entities of the same type, for example all students. In an ER diagram an entity set is drawn as a rectangle, and each entity is described by its attributes.

2. Attributes

Attributes are the properties that describe an entity. A simple attribute holds a single indivisible value, such as age. A composite attribute can be divided into smaller parts, such as a name made of first name and last name. A multivalued attribute can hold more than one value, such as several phone numbers. A derived attribute can be calculated from another attribute, such as age from date of birth.

3. Relationship

A relationship is an association between two or more entities, for example a student enrols in a course. A relationship set is the collection of such associations, and it is drawn as a diamond in an ER diagram. A relationship can have its own attributes, such as the enrolment date or the grade. Most relationships are binary, which means they connect two entity sets.

4. Cardinality

Cardinality describes how many entities in one entity set can be associated with entities in another. In a one-to-one relationship, each entity is linked to at most one entity on the other side, for example a person and a passport. In a one-to-many relationship, one entity is linked to many on the other side, for example one lecturer teaches many courses. In a many-to-many relationship, entities on both sides can be linked to many, for example students and courses. A many-to-many relationship is implemented with a separate linking table.

5. Keys in an ER Model

A primary key is an attribute, or a set of attributes, that uniquely identifies each entity in an entity set, and it is shown underlined in an ER diagram. A candidate key is any attribute set that could serve as the primary key. A foreign key is an attribute in one table that refers to the primary key of another table. When an ER diagram is converted into tables, foreign keys are how relationships are represented.

6. Weak Entity

A weak entity is an entity that does not have a key of its own that is enough to identify it. It depends on a strong entity, called the owner, and is identified by combining its own partial key, called the discriminator, with the primary key of the owner. A weak entity always takes part totally in its identifying relationship, and it is drawn as a double rectangle. An example is a dependent of an employee, or a room inside a building.